University of Florida, Department of Mathematics

Algebra First-Year Exam, August 2023, Part 1

Answer four problems. Write your answers clearly in complete English sentences.

1. Suppose G is a group and H, K are normal subgroups of G such that $H \cap K = 1$.

(i) Show that there is an injective homomorphism $G \to G/H \times G/K$.

(ii) Show that if $G = HK$ then G is isomorphic to $G/H \times G/K$.

2. Suppose G is a finite group, p is a prime, $|G|$ is a power of p and $G \times S \to S$ is an action of G on S. Show that if p does not divide $|S|$ then S has an element fixed by every element of G.

3. Suppose p and q are primes such that $p < q$ and $p \mid q-1$. Show that there is a unique nonabelian group of order pq, and show it is a semidirect product of two cyclic groups.

4. Give the definitions of

(i) solvable group

(ii) nilpotent group.

(iii) Show that A_4 is solvable but not nilpotent.

5. Show that if $n \geq 5$ then A_n has no proper subgroup of index less than n (you may assume results about the simplicity of A_n).

6. In both cases, justify your answer.

(1) Find all conjugacy classes of elements of order 4 in S_6.

(2) Do the same for A_6.